

CDE 2300 **Product Design and Innovation**

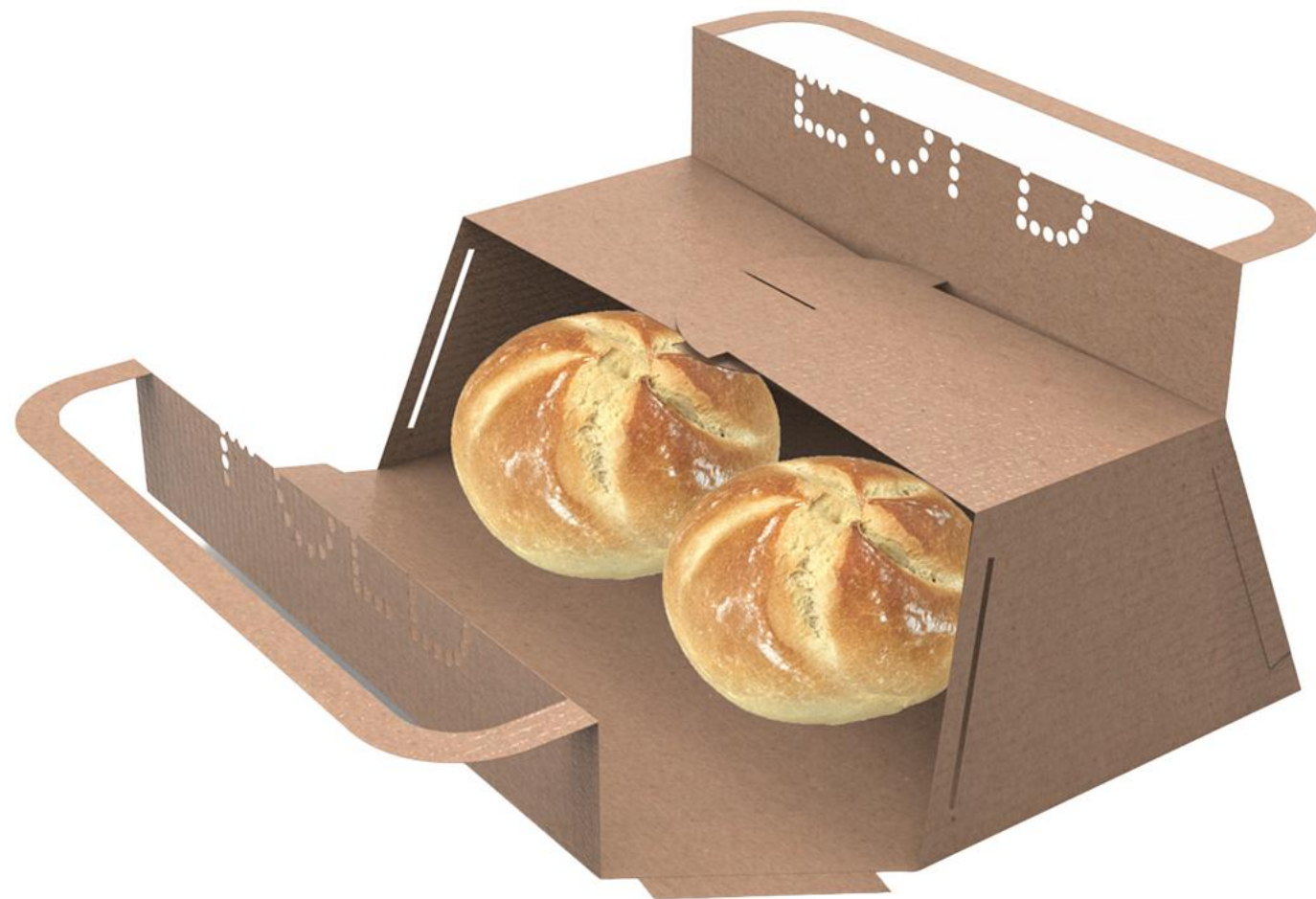
Hi im Darren

Y2 DiD Student



FOLD

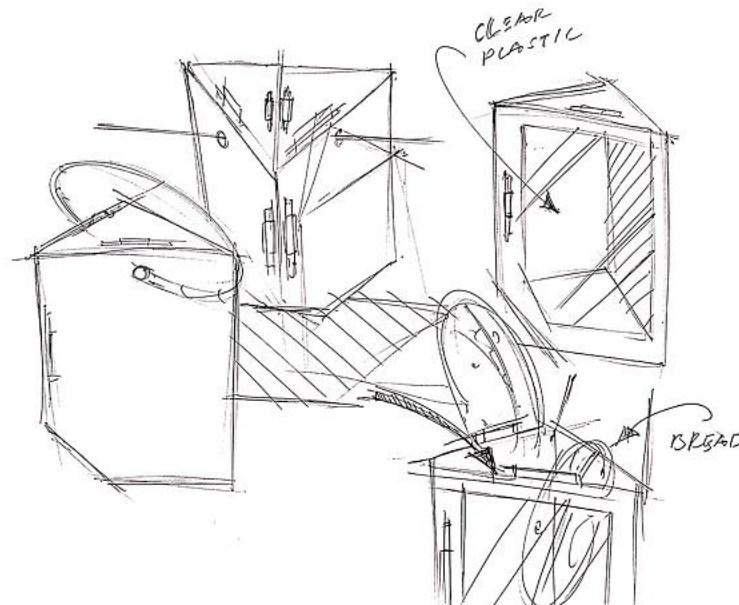
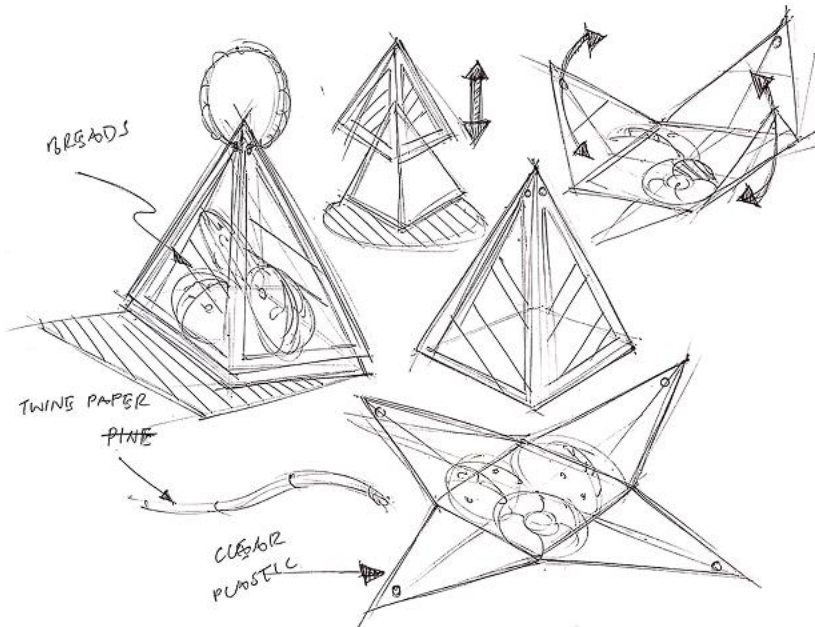
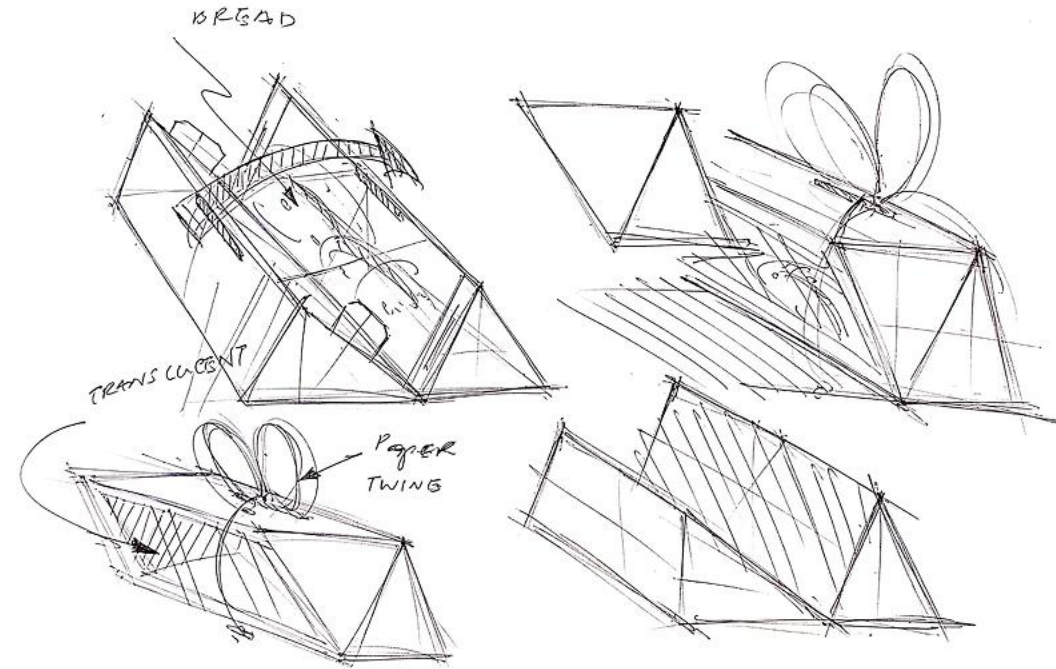
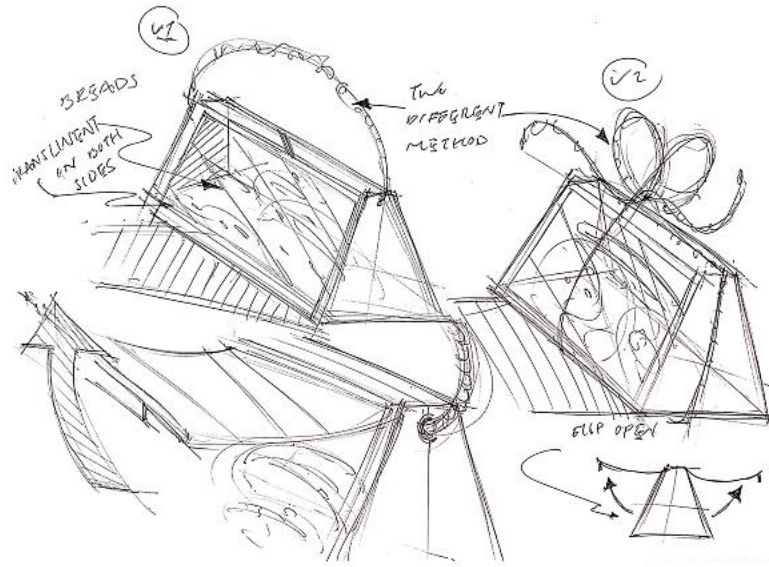
Fold is a recyclable bread packaging that replaces single-use plastic with a simple paper fold reducing waste while turning everyday purchase into a small, thoughtful ritual.



How might we **replace single-use plastic for
takeaway bread with a recyclable solution that
**transforms every opening into a quietly
delightful ritual?****

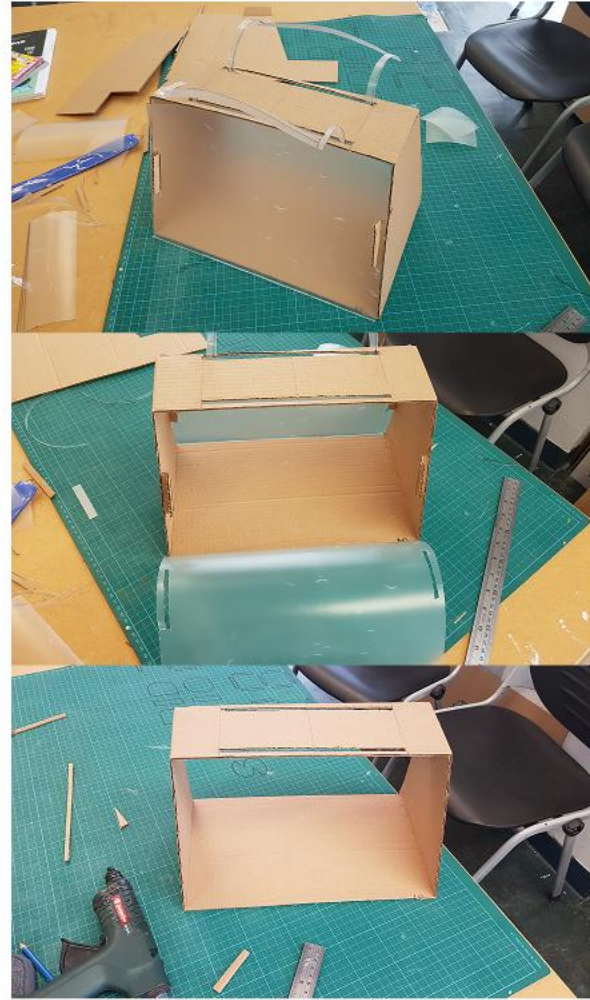
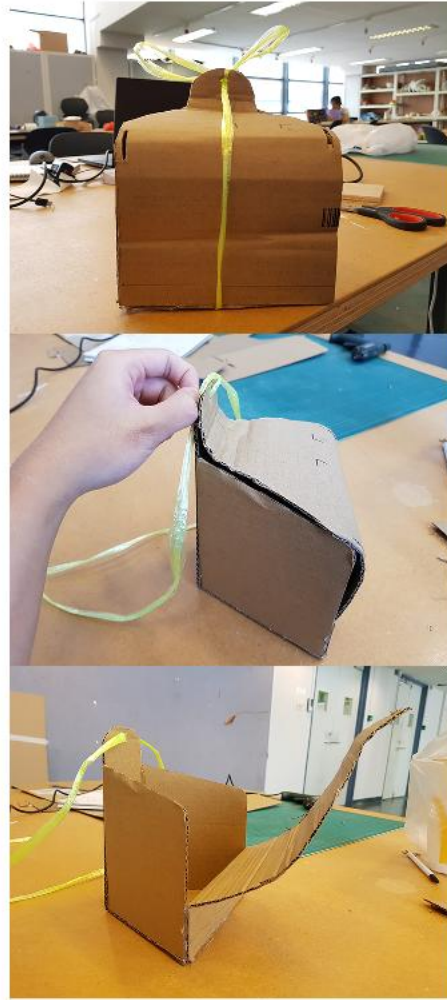
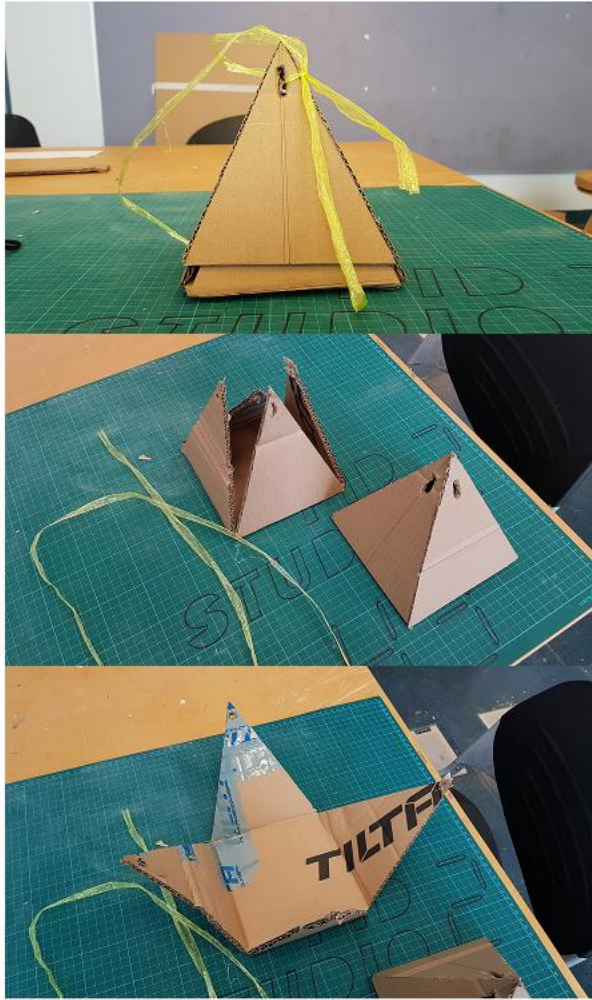


Finding Ideas



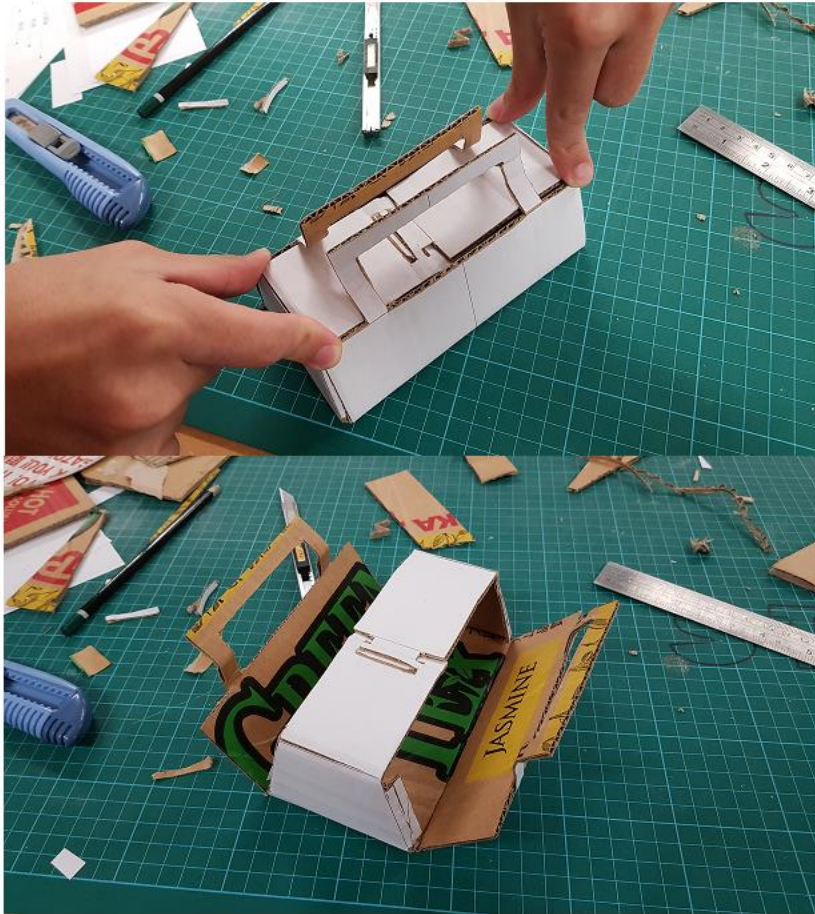
Quick Ideation Sketches

Sketching out initial ideas, can be wild & without much constraints.



Low Fidelity Prototype

Quick and dirty prototype to figure out the proportion and visibility.



Mid Fidelity Prototype

Iteration mock-up from chosen concept.

Numbers of iteration mock-up depends on the nature of the project.

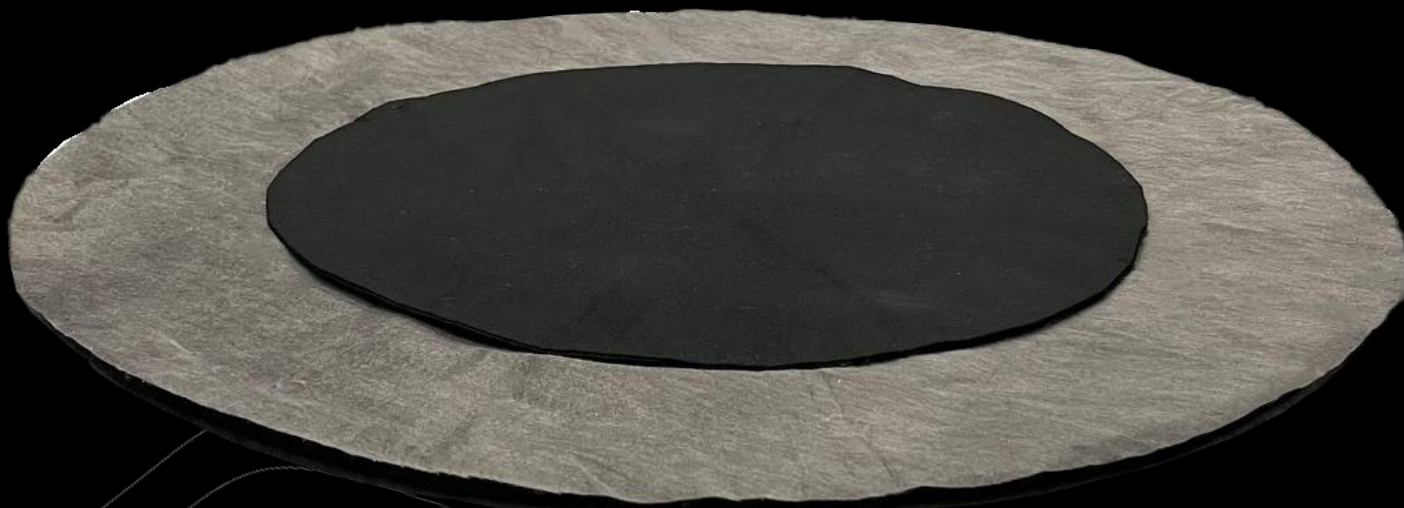
Hello!

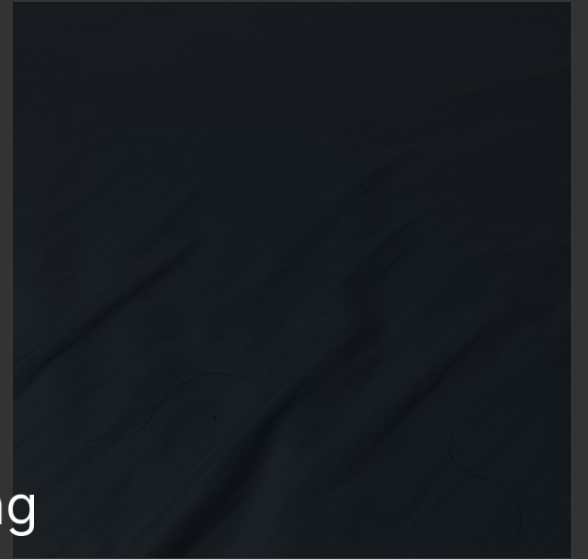
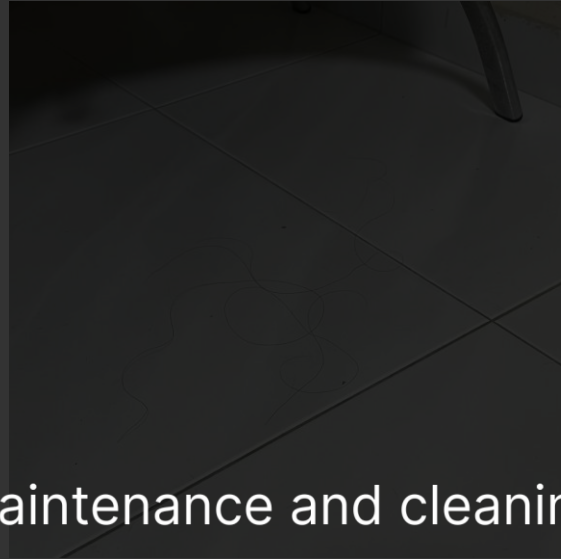
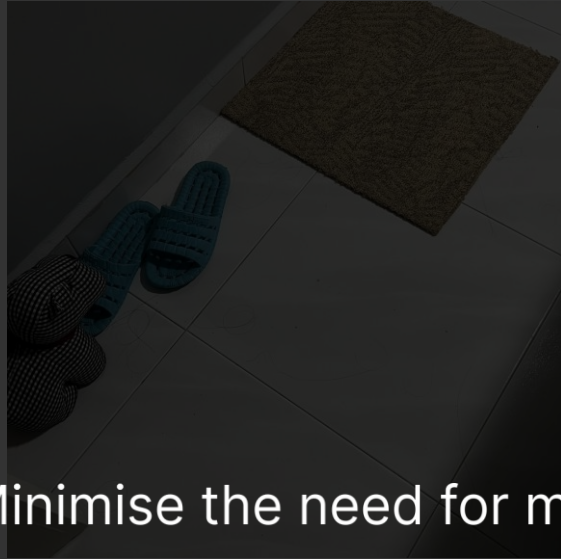
My name is Bo Ee!
Year 2 Industrial Design



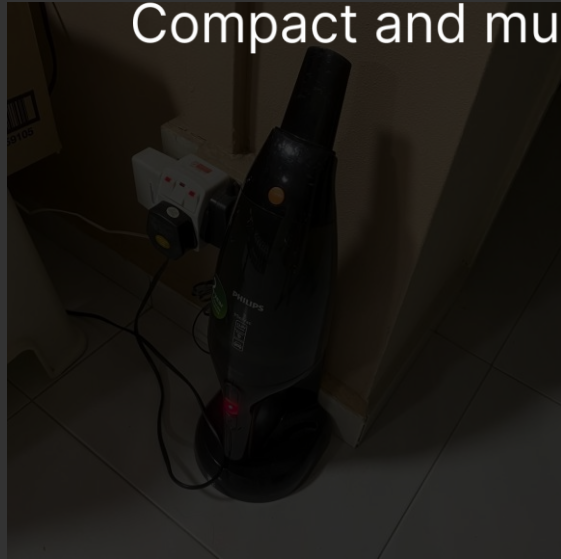
(c)hairt

A concept of a floor mat that collects hair as you go with electrostatic sheets. Designed to streamline the whole process of cleaning up shed hair.





Minimise the need for maintenance and cleaning



Compact and multi-purpose design

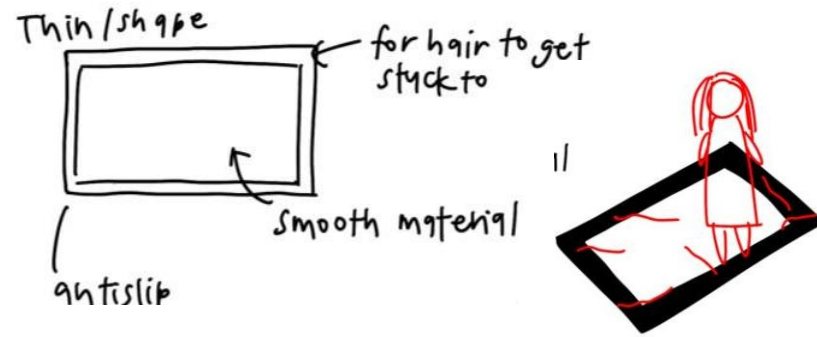
A photograph showing a person's bare feet standing on a dark, circular mat placed on a light-colored tiled floor. The mat has a lighter, circular center and a darker, irregular outer ring. The person's legs are visible from the knees down. The floor is made of large, light-colored square tiles.

“ How might we design and create a product/ service for family members doing household chores, that would contribute to effective clean up of hair and dust in areas of the homes, so that the entire cleaning process for users will be faster and more convenient?

”

Concept 3: Hair collecting floor mat

Is it the sight of hair that leads user to start cleaning?
What if we can hide hair from users?



Exploring shapes :

● - electrostatic sheet

● - silk fabric

● - considerations

①



- longer length might make it hard to replace ES sheet
- good to have a large area for users to walk over so they won't keep stepping on hair.

②



- easier when it's a circle as shape (?) is consistent
- not sure if users would prefer a round floor mat (find out through user testing)





REPLACABLE SHEETS

With just a few easy steps



CDE 2300 **Product Design and Innovation**

Innovation is not invention

Don't need a new law of physics or an amazing breakthrough



Invention: Creating a new technology or process that didn't exist before (e.g., the first laser).



Innovation: Creating new value for a user (e.g., using a laser to scan groceries).

You don't need a PhD in Physics to be an innovator; you need a "PhD" in your user's daily frustrations.

The Problem: Glass bottles
were iconic but frustrating.
The "last 15%" of the ketchup
was trapped, and the flow was
hard to control.

Successful innovations come from
reconfiguring what already exists to
better serve human needs

Design driven innovation

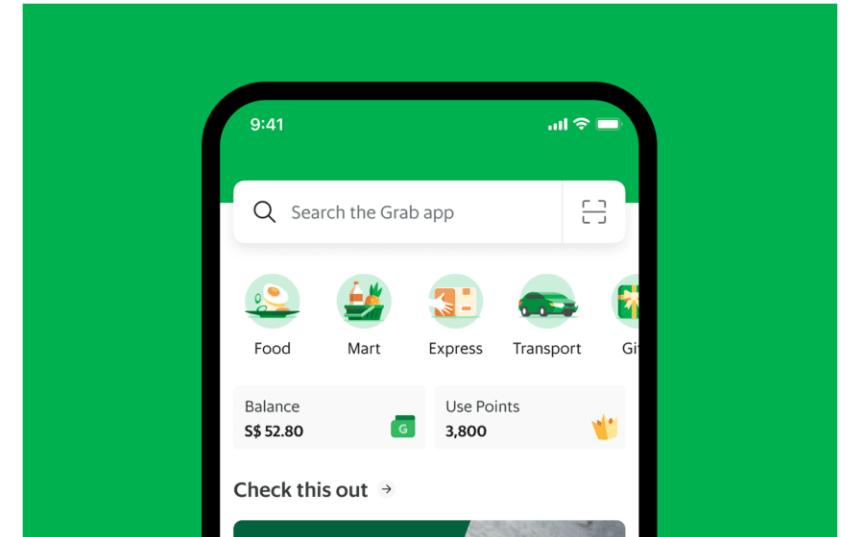
Switched to squeezable plastic
Flipped the bottle upside down
Used a simple silicone valve to prevent leaks.

Increased consumption and massive customer satisfaction



Design driven innovation

Friction Removal: Finding the "annoying" parts of a task and designing them away.



How to Innovate with Design

Empathize: Watch a user struggle with a "normal" product.

Define: Identify the "Friction Point" (e.g., "The handle is too thin," "The buttons are too small")

Frame: Jobs to be done (JTBD)

Ideate: How can we fix this using materials we already have?

Prototype: Build a low-cost version and test it immediately.

Ideate: How can we fix this using materials we already have?



Humane Ai Pin

Take Ai with you everywhere.



The Mistake: Humane designed a cool gadget first, then tried to find a reason for it to exist.

The Reality: Users didn't have a need to talk to a pin on their chest. The "job" of checking a notification was already being done better, faster, and more privately by the smartphone or smartwatch.

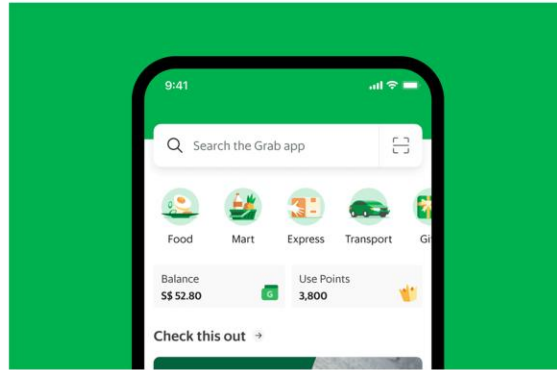
Why This Matters for the Future: In a world where technology is becoming a "commodity" (everyone has access to the same software and hardware), design is the only remaining differentiator.

The one that wins is the one that designed a better experience.



CDE 2300 Product Design and Innovation

Digital | Untangible



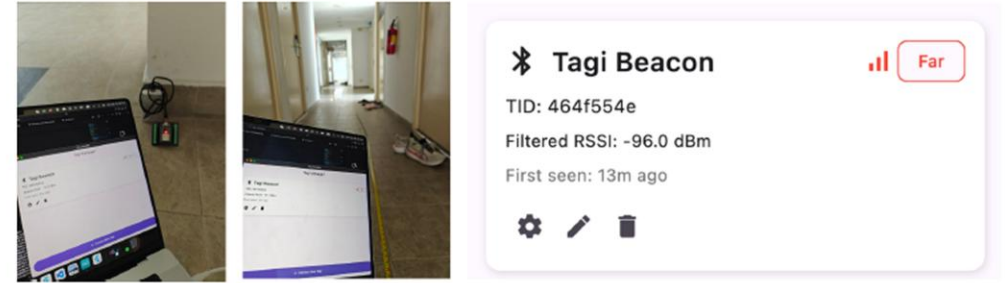
Physical | Tangible



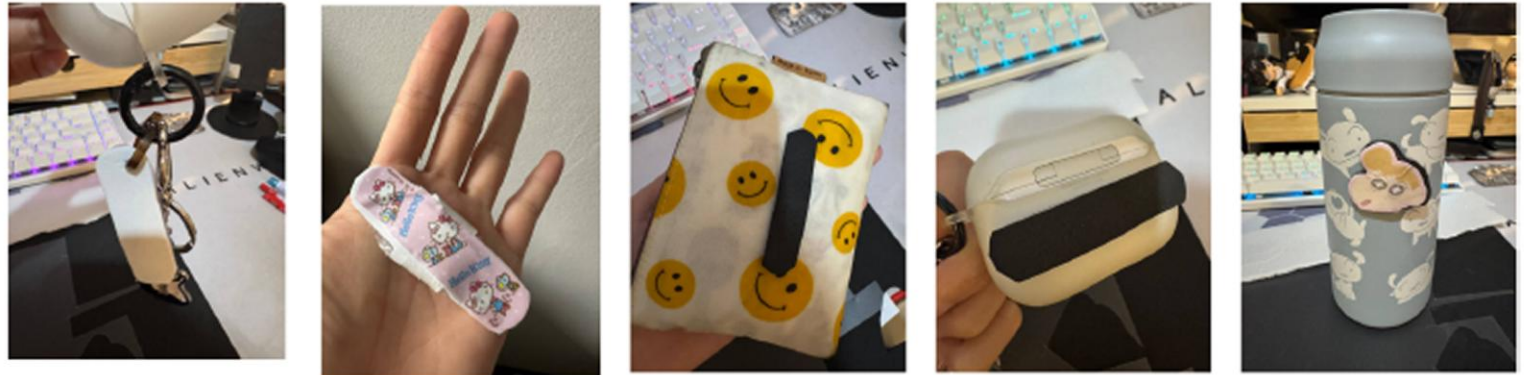
Tagi

a **flexible, sticker-style device** that could be easily attached to almost any item to track it without active supervision

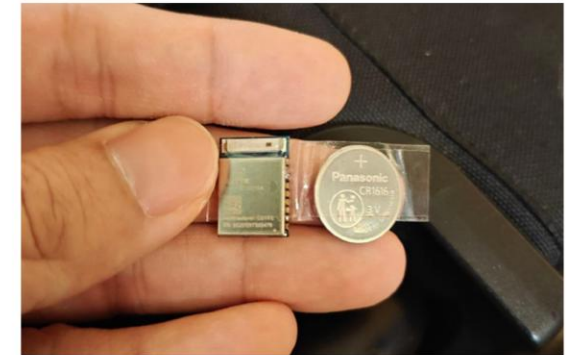
New format for tracking small but important items in our everyday



Tagi's Functional Testing



"When I'm moving between places, I want **proactive alerts** if I've left anything behind, so I can avoid losing items and maintain peace of mind when going about."



Actual Expected Size of Functional Prototype

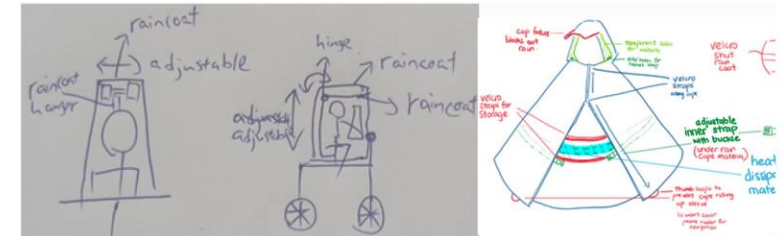
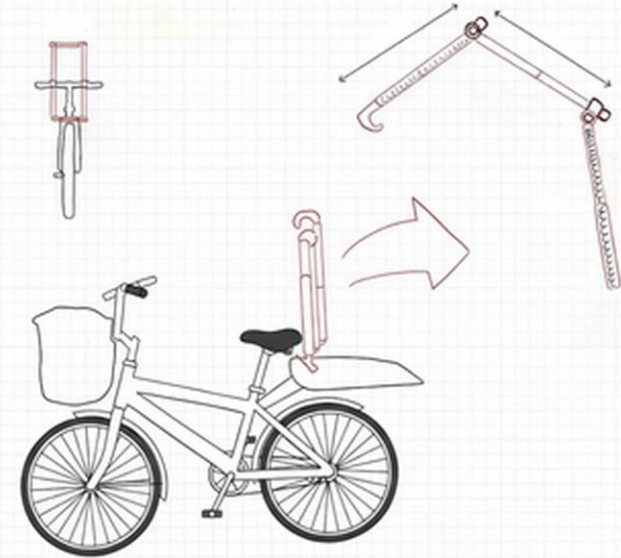
Cytect

a bike-mounted deployable weather canopy system for cyclists on the go

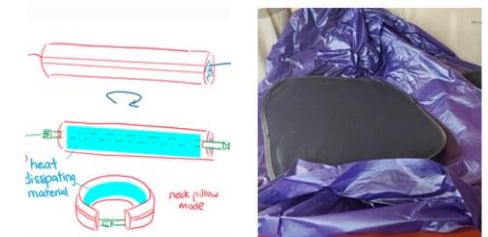
Poncho system reimaged and integrated into a bicycle for cyclists



Bicycle-Mounted Deployable Weather Canopy - Prototype Design Sheet



"We recognize that putting on the poncho was a tedious task for cyclists on the go, therefore we came up with a solution to streamline this process of "sheltering" on the go."



"Most people make the mistake of thinking design is what it looks like... That's not what we think design is. It's not just what it looks like and feels like. Design is how it works."

Steve Jobs

If you need help!

Luke - @notwatermelon

Bo ee -@starboee

Darren - @Derrickkkkkkkkkkkkkkkkkkkkk

Prerana - @preranaac

Erica - @ericamolenberg

Join our telegram group

Join our Instagram page



@cde2300pdi

Brief

Design something that solves a problem

Groups of 3 to 4 team members

National winner S\$8,400	International runners up S\$8,400
Sustainability winner S\$50,700	International winner S\$50,700

THE BRIEF

DESIGN SOMETHING THAT SOLVES A PROBLEM

The brief is broad. We're looking for designers who think differently, to create products that work better.

Engineers follow an iterative design process. The judges – and James Dyson especially – are drawn to designs that employ clever yet simple engineering principles and address clear problems.

As well as proving your project's technical viability, we'd also love to see that it's commercially viable, too – so include any research you've done into manufacturing costs and retail prices.

James Dyson will also look for entries that address a sustainability issue, or have been designed, sourced or manufactured sustainably.

[FIND OUT HOW TO ENTER](#)

"Young design engineers have the ability to develop tangible technologies that can change



James Dyson Award

Common topics to look at...

Inclusive & Accessible living

Example - Accessible sinks and mirrors
for wheelchair users

Enhancing Everyday Living

Example - Food waste reduction |
Improve the waking up experience

Home & Urban Environments

Example - Table-top fan redesign to
help cope with indoor heat

Identify and Address
(a problem) with
a tangible solution

Course Overview

Week	Titles	Week	Titles
1 (12 Jan)	Innovation-driven economy, Analogy and metaphor in design	7 (2 Mar)	Prioritization map, Concept selection, Specifications
2 (19 Jan)	JTBD, Job story and Job statement	8 (9 Mar)	Prototyping and Prototyping Plan
3 (26 Jan)	Users' goals, Product values and Product features	9 (16 Mar)	Storytelling
4 (2 Feb)	Product vision, Systematic ideation and Concept generation	10 (23 Mar)	Testing
5 (9 Feb)	Mid Term Review	11 (30 Mar)	Group consultation
6 (16 Feb)	No class!	12 (6 Apr)	Group consultation
23 Feb to 27 Feb	Recess Week	13 (13 Apr)	End Term Review (Show and Tell) (2pm to 6pm)

Assessment

Individual

- Peer Review (5%)
- Design Reflection (5%)
- Assignments (5%)

Group

- End Term Deliverables (73%)
- Mid Term Deliverables (12%)